

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

Name: Brune Boukobza

ID: BWB78

Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/Desktop/BEE 4750/hw5-bruneb`
```

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

The MRF recycling rate is 40%, and the ash fraction of non-recycled waste is 16% and of recycled waste is 14%. Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

```
##CODE TO PROBLEM 1.1

##Data from table
#Components
components = [
    "Food", "Paper", "Plastics", "Textiles", "Rubber", "Wood",
    "Yard", "Glass", "Ferrous", "Aluminum", "OtherMetal", "Misc"
]
#Percent total mass
mass_pct = [15, 40, 5, 3, 2, 5, 18, 4, 2, 2, 1, 3]
#Combustion ash
ash_pct = [8, 7, 5, 10, 15, 2, 2, 100, 100, 100, 100, 70]
#MRF recycling rate
recycle_pct = [0, 55, 15, 10, 0, 30, 40, 60, 75, 80, 50, 0]
#Convert to fractions
```

```

mass_frac = mass_pct ./ 100
ash_frac = ash_pct ./ 100
recycle_frac = recycle_pct ./ 100

#Recycling fraction
recycling = sum(mass_frac .* recycle_frac)
#Fraction of waste that actually becomes recycling residuals which is the material that goes
residual_frac = sum(mass_frac .* recycle_frac .* (1.- recycle_frac))
#Ash fraction of raw MSW going directly to WTE
ash_raw_WTE = sum(mass_frac .* ash_frac)
#Ash fraction of recycling residuals
ash_residuals = sum((mass_frac .* recycle_frac .* (1.- recycle_frac)) .* ash_frac) /
                sum(mass_frac .* recycle_frac .* (1.- recycle_frac))

println("Recycling fraction = ", round(recycling, digits=3))
println("Ash fraction of raw MSW sent to WTE = ", round(ash_raw_WTE, digits=3))
println("Ash fraction of recycling residuals = ", round(ash_residuals, digits=3))

```

```

Recycling fraction = 0.378
Ash fraction of raw MSW sent to WTE = 0.164
Ash fraction of recycling residuals = 0.153

```

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

a_{cf} : waste transported from city c to facility f . There are nine of these. b_{df} : leftover waste being transported from disposal facility d to f . There are 3 of these. c_f : binary variable to see if facility f is working.

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Minimize cost, so combine transportation and disposal. If land-fill=1, MRF=2, and WTE=3, then

Transportation costs: $1.5(5a_{11}+30a_{12}+15a_{13}+15a_{21}+25a_{22}+10a_{31}+13a_{32}+45a_{33}+20a_{41}+32b_{21}+18b_{31}+15b_{41})$

Disposal costs: LF $2000c_1+50(a_{11}+a_{21}+a_{31}+b_{21}+b_{31})$ MRF $1500c_2+7(a_{12}+a_{22}+a_{32})+0.38(40)(a_{12}+a_{22}+a_{32})$ WTE $2500c_3+60(a_{13}+a_{23}+a_{33}+b_{23})$

Now: $2000c_1 + 1500c_2 + 2500c_3 + 57.5a_{11} + 67.2a_{12} + 82.5a_{13} + 72.5a_{21} + 59.7a_{22} + 75a_{23} + 69.5a_{31} + 89.7a_{32} + 90a_{33} +$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

Constraints: mass balance, capacity, non-negativity, binary, and demand, where all waste from each city must be treated.

Mass balance: For ash mass balance, consider waste sent directly to WTE has an ash fraction of 0.164 and waste sent from recycling to WTE has an ash content of 0.14. $0.164(a_{13} + a_{23} + a_{33}) + 0.14(b_{23}) = b_{31}$

Recycling mass balance: $b_{21} + b_{23} = (1 - 0.38)(a_{12} + a_{22} + a_{32})$

Capacity: LF: $a_{11} + a_{21} + a_{31} + b_{21} + b_{31} \leq 200$ MRF: $a_{12} + a_{22} + a_{32} \leq 350$ WTE: $a_{13} + a_{23} + a_{33} + b_{23} \leq 210$

Demand: City one: $a_{11} + a_{12} + a_{13} = 100$ City two: $a_{12} + a_{22} + a_{32} = 90$ City three: $a_{31} + a_{32} + a_{33} = 120$

Binary: set $c_1 = 1$ because the landfill always has to operate. Use m: $a_{12} + a_{22} + a_{32} \leq m_2 c_2$; $m_2 = 350$ $a_{13} + a_{23} + a_{33} + b_{23} \leq m_3 c_3$, $m_3 = 210$

Non-negativity defines all variables as being ≥ 0

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
##CODE FOR PROBLEM 1.5

m2 = 350
m3 = 210
waste = Model(HiGHS.Optimizer)
@variable(waste, a[1:9] >= 0)
@variable(waste, b[1:3] >= 0)

#Binary variables
@variable(waste, c[1:3], Bin)
@constraint(waste, commit2, a[2] + a[5] + a[8] <= m2 * c[2])
@constraint(waste, commit3, a[3] + a[6] + a[9] + b[3] <= m3 * c[3])

#Landfill is always operating constraint
@constraint(waste, commit1, c[1] == 1)
```

```

#Cost
@objective(waste, Min, 2000*c[1]+1500*c[2]+2500*c[3]+57.5*a[1]+67.2*a[2]+82.5*a[3]+72.5*a[4]
+59.7*a[5]+75*a[6]+69.5*a[7]+89.7*a[8]+90*a[9]+98*b[1]+77*b[2]+82.5*b[3])

#Capacity constraints
@constraint(waste, ca1, (a[1]+a[4]+a[7]+sum(b[1:2])) <= 200))
@constraint(waste, ca2, (a[2]+a[5]+a[8] <= 350))
@constraint(waste, ca3, (a[3]+a[6]+a[9]+b[3] <= 210))
#Waste demand constraints
@constraint(waste, d1, (sum(a[1:3]) == 100))
@constraint(waste, d2, (sum(a[4:6]) == 90))
@constraint(waste, d3, (sum(a[7:9]) == 120))
#Mass balance constraints
@constraint(waste, MRF, (0.62*(a[2]+a[5]+a[8]) == b[1]+b[3]))
@constraint(waste, WTE, (0.164*(a[3]+a[6]+a[9])+0.14*(b[3]) == b[2]))

optimize!(waste)

display(value.(a))
display(value.(b))
display(value.(c))
cost = objective_value.(waste)
display(cost)

```

9-element Vector{Float64}:

```

100.0
 0.0
 0.0
-0.0
-0.0
 90.0
78.42105263157896
 0.0
41.578947368421034

```

3-element Vector{Float64}:

```

0.0
21.578947368421048
0.0

```

3-element Vector{Float64}:

1.0
-0.0
1.0

27853.94736842105

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms
MIP has 11 rows; 15 cols; 41 nonzeros; 3 integer variables (3 binary)

Coefficient ranges:

Matrix [1e-01, 4e+02]
Cost [6e+01, 2e+03]
Bound [1e+00, 1e+00]
RHS [1e+00, 4e+02]

Presolving model

9 rows, 14 cols, 37 nonzeros 0s

8 rows, 13 cols, 35 nonzeros 0s

Presolve reductions: rows 8(-3); columns 13(-2); nonzeros 35(-6)

Solving MIP model with:

8 rows

13 cols (2 binary, 0 integer, 0 implied int., 11 continuous, 0 domain fixed)

35 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;

I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized round;

S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution;

Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial upper bound;

	Nodes		B&B Tree		Objective Bounds			Dynamic Cuts	
Src	Proc.	InQueue	Leaves	Expl.	BestBound	BestSol	Gap	Cuts	InQueue
J	0	0	0	0.00%	-inf	46026.194839	Large	0	
S	0	0	0	0.00%	2000	27853.947368	92.82%	0	

50.0% inactive integer columns, restarting

Model after restart has 1 row, 3 cols (0 bin., 0 int., 0 impl., 3 cont., 0 dom.fix.), and 3 nonzeros

0	0	0	0.00%	26920.363409	27853.947368	3.35%	0
1	0	1	100.00%	27853.947368	27853.947368	0.00%	0

Solving report

Status Optimal

Primal bound	27853.9473684
Dual bound	27853.9473684
Gap	0% (tolerance: 0.01%)
P-D integral	0.000574344766147
Solution status	feasible
	27853.9473684 (objective)
	0 (bound viol.)
	0 (int. viol.)
	0 (row viol.)
Timing	0.00
Max sub-MIP depth	0
Nodes	1
Repair LPs	0
LP iterations	6
	0 (strong br.)
	0 (separation)
	0 (heuristics)

The minimum objective is \$27,793 per day.

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

It is best to not use the recycling facility because the transportation costs are high since the MRF facility is far from the three cities.

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500. In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Attached to second pdf form

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

```
import Pkg
Pkg.add("CSV")
using CSV

#Load data
genfile = "data/generators.csv"
df = CSV.read(genfile, DataFrame)

plants = df.Plant
Pmin = Dict{df.Plant => df.Pmin}
Pmax = Dict{df.Plant => df.Pmax}
VarCost = Dict{df.Plant => df.VarCost}
Ramp = Dict{df.Plant => df.Ramp}

#Renewable generators
is_wind = df.Resource .== "Wind"
is_solar = df.Resource .== "Solar"

#Generator sets
thermal = plants[.!is_wind .& .!is_solar]
wind = plants[is_wind]
solar = plants[is_solar]

##Scenario Tree Data
#Period 1
d1 = 1100
cf_solar_t1 = 0.9
cf_wind_t1 = 0.45
#Period 2
scenarios = [
    (name="S1", d=1200, cf_s=0.95, cf_w=0.40, p=0.75*0.70),
    (name="S2", d=1200, cf_s=0.75, cf_w=0.50, p=0.75*0.30),
```

```

        (name="S3", d=1500, cf_s=0.95, cf_w=0.40, p=0.25*0.70),
        (name="S4", d=1500, cf_s=0.75, cf_w=0.50, p=0.25*0.30),
    ]
    S = length(scenarios)
    model = Model(HiGHS.Optimizer)
    #Period 1 generation
    @variables(model, begin
        g1[p in plants] >= 0
    end)
    #Period 2 generation
    @variables(model, begin
        g2[p in plants, s in 1:S] >= 0
    end)

    ##Constraints
    #Period 1
    for p in plants
        @constraint(model, Pmin[p] <= g1[p] <= Pmax[p])
    end
    #Renewables in period 1 have CF limits that we must account for
    for p in wind
        @constraint(model, g1[p] <= cf_wind_t1 * Pmax[p])
    end
    for p in solar
        @constraint(model, g1[p] <= cf_solar_t1 * Pmax[p])
    end
    #Period 1 energy balance
    @constraint(model, sum(g1[p] for p in plants) == d1)

    #Period 2
    for (s, sc) in enumerate(scenarios)
        #Generator limits
        for p in plants
            @constraint(model, Pmin[p] <= g2[p,s] <= Pmax[p])
        end
        #Renewable CF limits
        for p in wind
            @constraint(model, g2[p,s] <= sc.cf_w * Pmax[p])
        end
        for p in solar
            @constraint(model, g2[p,s] <= sc.cf_s * Pmax[p])
        end
    end

```

```

    for p in plants
        @constraint(model, g2[p,s] - g1[p] <= Ramp[p])
        @constraint(model, g1[p] - g2[p,s] <= Ramp[p])
    end
    #Energy balance each scenario
    @constraint(model, sum(g2[p,s] for p in plants) == sc.d)
end

##Minimize Cost
@expression(model,
    cost1, sum(VarCost[p] * g1[p] for p in plants)
)
@expression(model,
    cost2,
    sum(scenarios[s].p * sum(VarCost[p] * g2[p,s] for p in plants)
        for s in 1:S)
)
@objective(model, Min, cost1 + cost2)

#Optimize
optimize!(model)

println("Optimal objective = ", objective_value(model))
println("\nPeriod 1 generation:")
for p in plants
    println(p, ": ", value(g1[p]))
end

println("\nPeriod 2 generation by scenario:")
for (s, sc) in enumerate(scenarios)
    println("\nScenario $(sc.name): p=$(sc.p)")
    for p in plants
        println("  ", p, ": ", value(g2[p,s]))
    end
end
end

```

```

Resolving package versions...
No Changes to `~/Desktop/BEE 4750/hw5-bruneb/Project.toml`
No Changes to `~/Desktop/BEE 4750/hw5-bruneb/Manifest.toml`

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 106 rows; 35 cols; 192 nonzeros

Coefficient ranges:

Matrix	[1e+00, 1e+00]
Cost	[4e-01, 4e+02]
Bound	[0e+00, 0e+00]
RHS	[1e+02, 2e+03]

Presolving model

45 rows, 30 cols, 94 nonzeros 0s

9 rows, 20 cols, 28 nonzeros 0s

Dependent equations search running on 2 equations with time limit of 1000.00s

Dependent equations search removed 0 rows and 0 nonzeros in 0.00s (limit = 1000.00s)

6 rows, 11 cols, 16 nonzeros 0s

Presolve reductions: rows 6(-100); columns 11(-24); nonzeros 16(-176)

Solving the presolved LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities	num(sum)
0	1.7720232374e+04	Pr: 2(1090)	0s
2	1.7926750000e+04	Pr: 0(0)	0s

Performed postsolve

Solving the original LP from the solution after postsolve

Model status : Optimal

Simplex iterations: 2

Objective value : 1.7926750000e+04

P-D objective error : 0.0000000000e+00

HiGHS run time : 0.00

Optimal objective = 17926.75

Warning: thread = 1 warning: parsed expected 6 columns, but didn't reach end of line around

@ CSV /Users/bruneboukoba/.julia/packages/CSV/XLcQT/src/file.jl:593

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 4. Filling remaining

@ CSV /Users/bruneboukoba/.julia/packages/CSV/XLcQT/src/file.jl:592

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 5. Filling remaining

@ CSV /Users/bruneboukoba/.julia/packages/CSV/XLcQT/src/file.jl:592

Warning: thread = 1 warning: only found 6 / 7 columns around data row: 6. Filling remaining

@ CSV /Users/bruneboukoba/.julia/packages/CSV/XLcQT/src/file.jl:592

Period 1 generation:

Biomass: 0.0

Hydroelectric: 195.0
Geothermal: 0.0
NG CCGT: 220.0
NG CT: 100.0
Wind: 135.0
Solar: 450.0

Period 2 generation by scenario:

Scenario S1: $p=0.5249999999999999$

Biomass: 0.0
Hydroelectric: 285.0
Geothermal: 0.0
NG CCGT: 220.0
NG CT: 100.0
Wind: 120.0
Solar: 475.0

Scenario S2: $p=0.22499999999999998$

Biomass: 0.0
Hydroelectric: 355.0
Geothermal: 0.0
NG CCGT: 220.0
NG CT: 100.0
Wind: 150.0
Solar: 375.0

Scenario S3: $p=0.175$

Biomass: 85.0
Hydroelectric: 500.0
Geothermal: 0.0
NG CCGT: 220.0
NG CT: 100.0
Wind: 120.0
Solar: 475.0

Scenario S4: $p=0.075$

Biomass: 100.0
Hydroelectric: 500.0
Geothermal: 0.0
NG CCGT: 275.0
NG CT: 100.0
Wind: 150.0

Solar: 375.0

References

List any external references consulted, including classmates.

Holly Archer